

E-Commerce Customer Behavior Analysis

Customer Analytics | Functional Analytics Project

Team Members

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Business Objective

Use analytics to understand e-commerce customer behavior and support better decisions on conversion, retention, and revenue.

1

Where do customers drop off?

Identify the stages — browsing, cart, checkout — that leak the most customers.

2

Which channels convert best?

Compare marketing channels and devices on conversion, not just traffic.

3

Who are the valuable customers?

Segment customers by recency, frequency, and monetary value (RFM).

Business Problem

Analyze session-level e-commerce data to deliver actionable insights that improve conversion, reduce cart abandonment, and grow customer lifetime value.

25,000

sessions analyzed

8,442

unique customers

5

customer segments (RFM)

Scope covers the full purchase funnel — from first visit to repeat purchase.

Dataset Overview & Tools

25,000

Rows / Sessions

29

Cleaned Columns

0

Missing / Duplicates

Dataset Fields

- Session-level e-commerce log: customer, device, marketing channel, product, pricing
- Behavioral flags: pages viewed, time on site, added to cart, purchased, cart abandoned
- Outcome fields: revenue, quantity, discount, rating

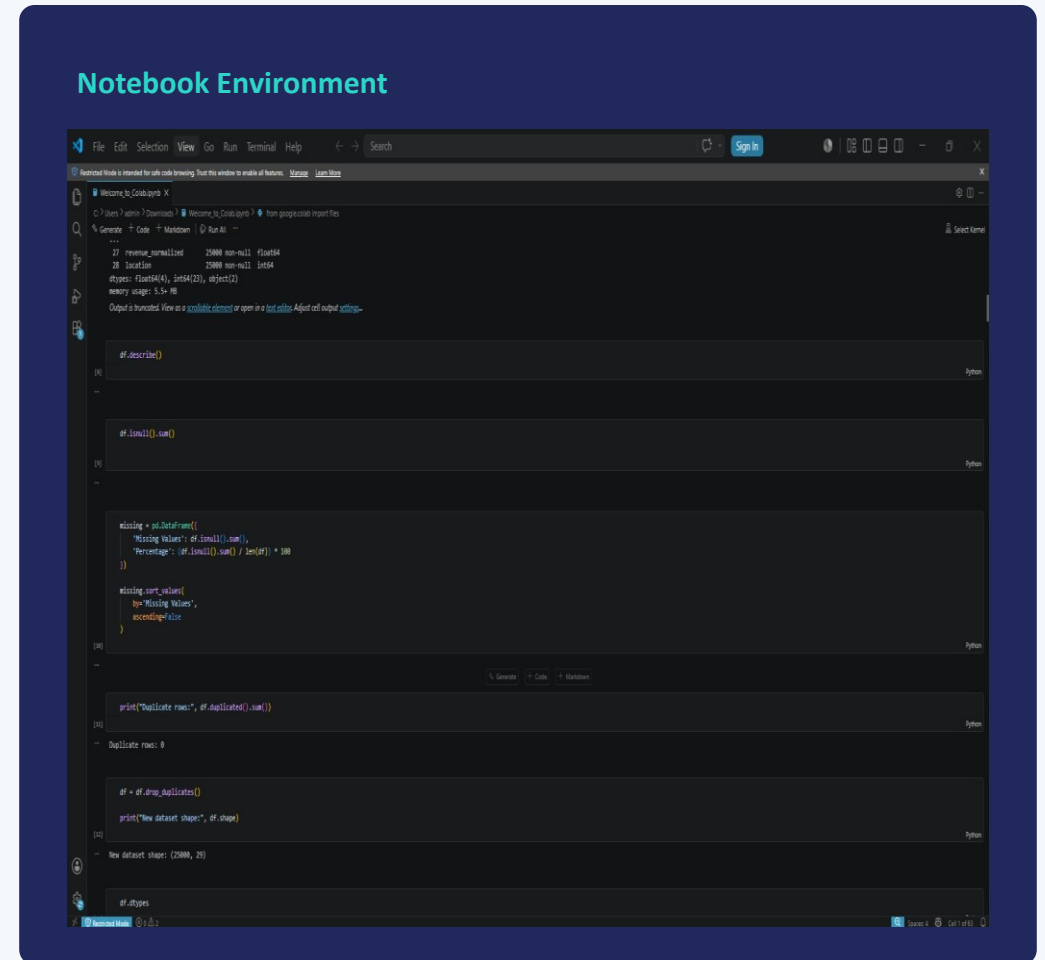
Tools & Techniques

Python (pandas, numpy)

Matplotlib & Seaborn

Power BI (DAX)

EDA & RFM



Steps Performed

The end-to-end workflow followed for this project

1

Data Collection

Loaded the 25,000-row e-commerce session dataset (Ecommerce.csv) into Python via pandas.

2

Data Cleaning

Checked shape, dtypes, and summary stats; confirmed zero missing values and zero duplicate rows.

3

Exploratory Data Analysis

Analyzed conversion, add-to-cart, and cart abandonment rates; plotted revenue by category, device, and channel.

4

Correlation Analysis

Built a correlation heatmap to see how cart activity, discounts, and revenue relate to purchase behavior.

5

Customer Segmentation (RFM)

Scored every customer on Recency, Frequency & Monetary value and grouped them into 5 segments.

6

Dashboard Building

Modeled the cleaned data in Power BI and built DAX measures (Total Revenue, Total Customers, Cart Abandoned).

7

Insight Generation

Translated the numbers into business recommendations on retention, checkout friction, and category focus.

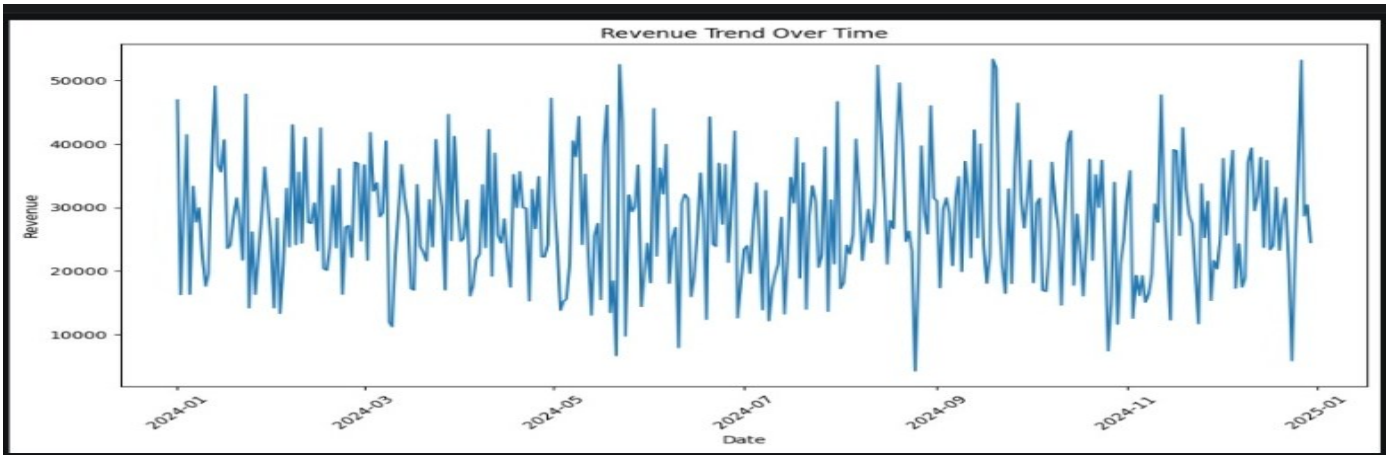
Key Performance Indicators

Overall funnel health across 25,000 sessions



Behavior & Revenue Trends

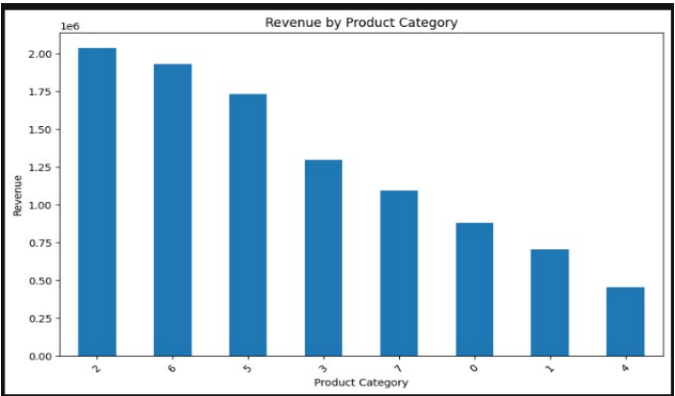
Daily Revenue Trend Over Time



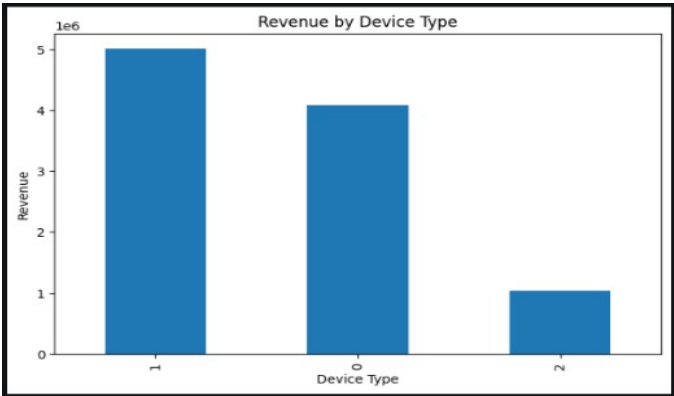
Reading the Trend

- Daily revenue swings between ~₹5K and ₹53K with no strong seasonal trend visible.
- Product Category 2 and Category 6 generate the highest revenue share.
- Device type 1 (likely desktop) drives ~5M in revenue vs 1M for device type 2.

Revenue by Product Category



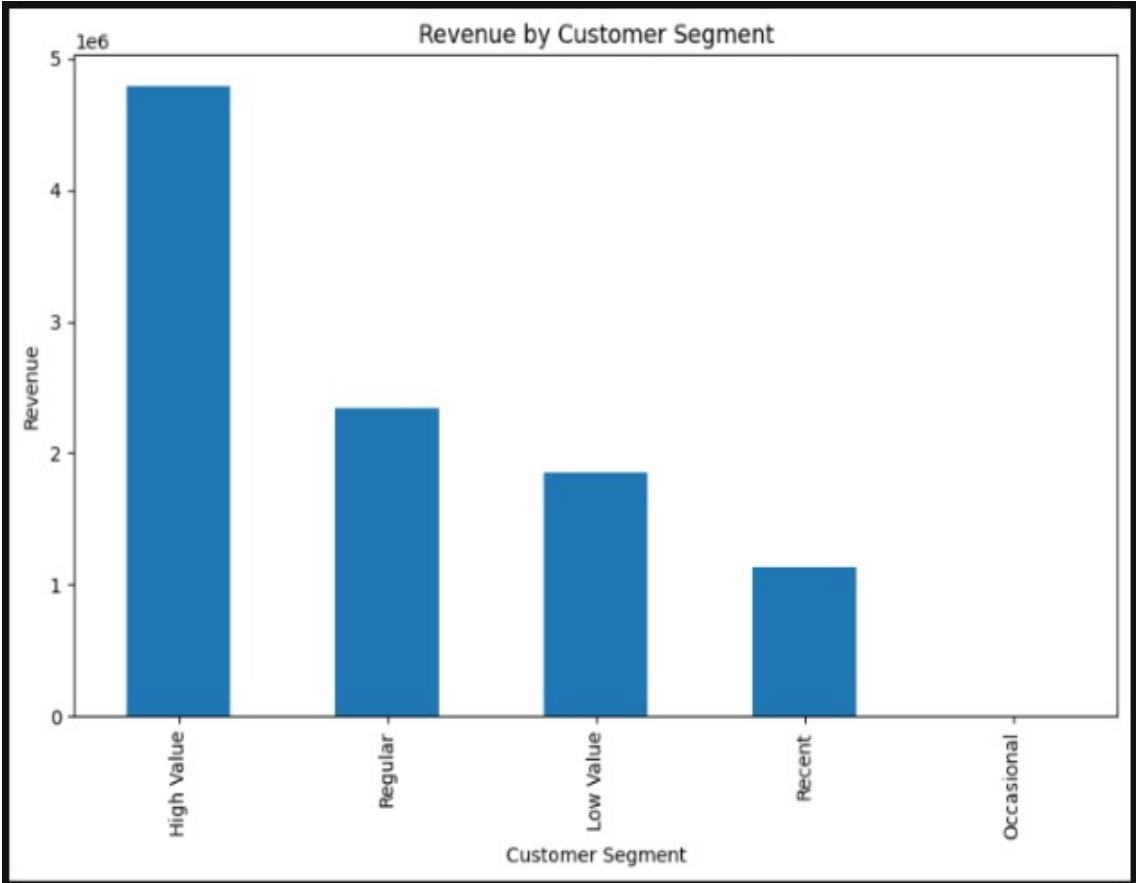
Revenue by Device Type



Customer Segmentation — RFM Analysis

Customers scored on Recency, Frequency & Monetary value, then grouped into actionable segments.

Revenue Contribution by Segment



High Value

R, F, M all ≥ 3 — most loyal, most profitable

Regular

Frequent & high spend, but not most recent

Recent

Just visited, spend/frequency not yet proven

Low Value

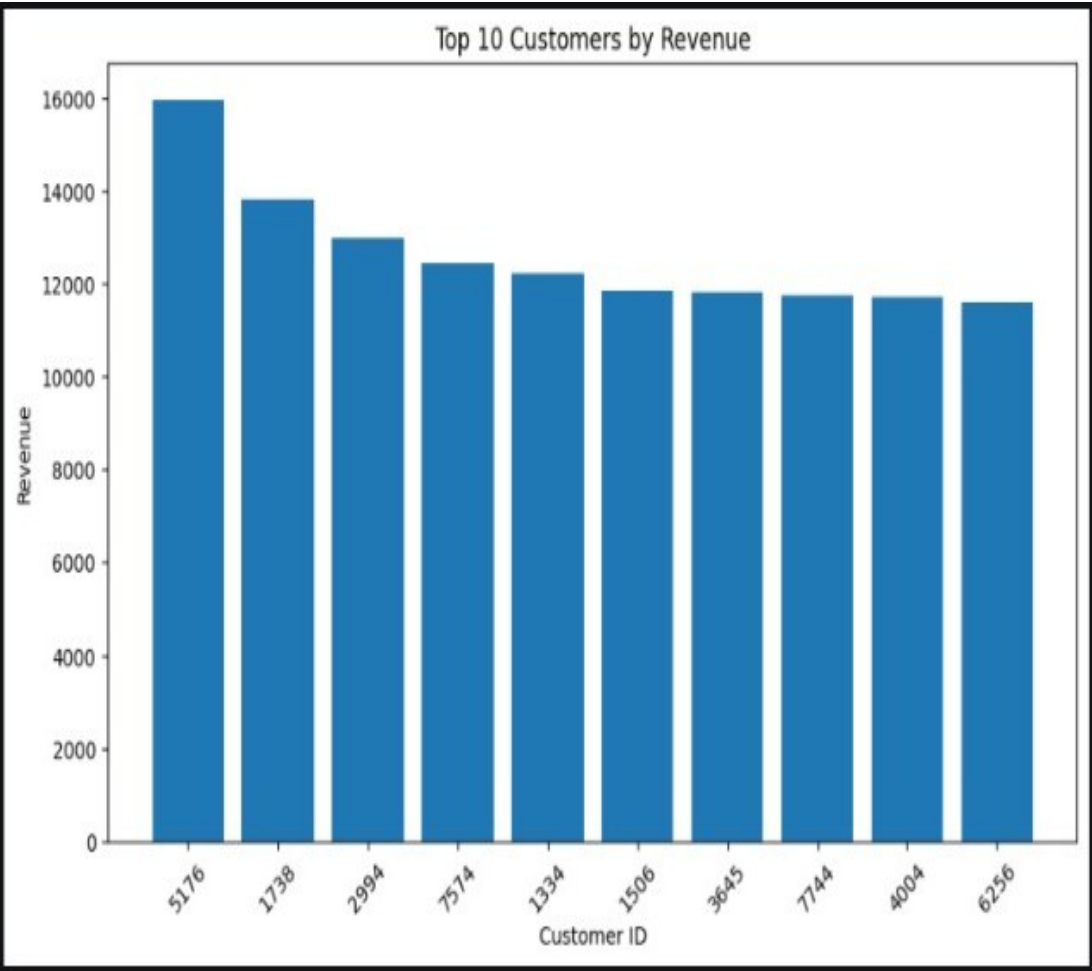
R and F both low — at risk / disengaged

Occasional

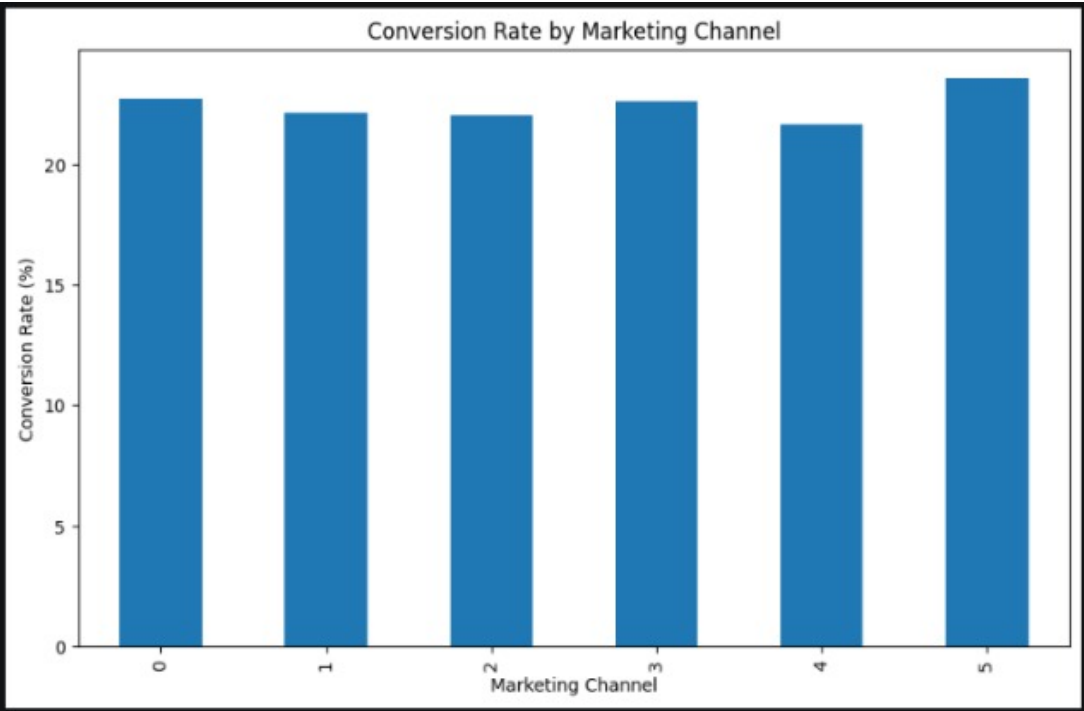
Everything else — irregular buyers

Purchase Behavior Patterns

Top 10 Customers by Revenue



Conversion Rate by Marketing Channel



All six channels convert in a tight 21.7%–23.6% band — no single channel underperforms badly, suggesting the funnel bottleneck sits after acquisition (at cart/checkout), not at channel targeting.

Power BI Executive Dashboard

Live measures: Total Revenue, Total Customers, Total Sessions, Cart Abandonment (DAX)



Insights & Recommendations

1

42% cart abandonment

Nearly half of carts never convert. A/B test checkout friction (extra fields, shipping surprises, forced login).

2

82% of customers are repeat buyers

6,918 of 8,442 customers return. Retention infrastructure is working — protect it with loyalty perks for the High Value segment.

3

Revenue concentrated in 2 categories

Category 2 and 6 drive the top revenue share — prioritize inventory and promotion budget there.

4

Channels convert almost equally

No channel underperforms; the leak is post-click, not pre-click. Fix the cart/checkout experience first.